

RTCG Pictorial Example

First Author Name
for initial Submission
City, Country
e-mail address

Second Author Name
Affiliation
City, Country
e-mail address

Third Author Name
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e-mail address

ABSTRACT

Shortly explain what you will be presenting as part of this pictorial. You can add the abstract after everything else is done. It should explain what kind of model you have chosen, which engine you are using, what exercises you decided to create for each of the micro-hand-ins. Not more than a sentence for each. All in all around 150- 200 words.

Author Keywords

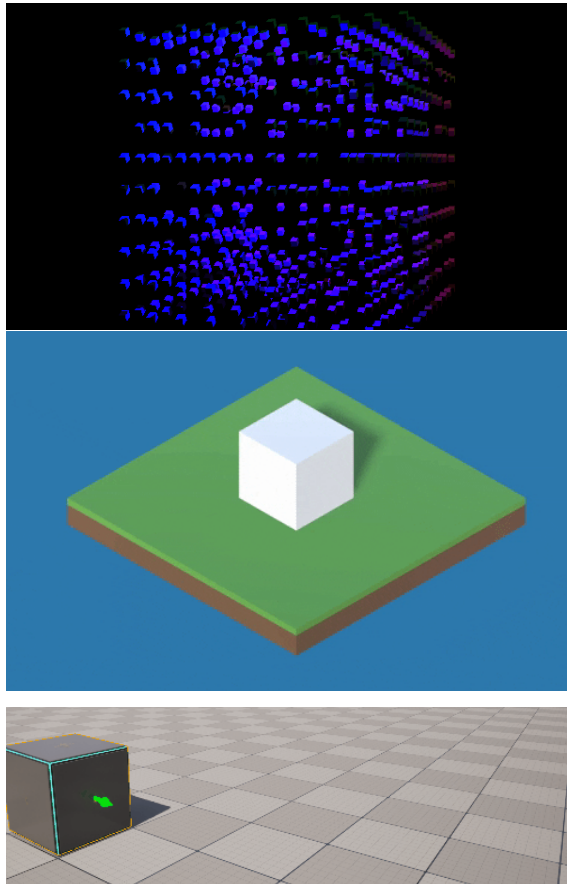
Keywords about your mini-project separated with semicolons, again focus on what you used and created with 1 word each.

FIRST BLOCK – GAME ENGINE

{Aba Help here does it make sense???? Maybe some example for a Unreal small project for the Unreal lecture the same as in the Unity lecture}

Show your final exercise visualization for **Unity and Unreal**, as a gif or as a series of images. If you have added an interesting configuration of lights, show a schematic of it, if you experimented with some additional code, add a simple pseudocode or code segment here. If you have made an interaction show how it works here, show a side by side of the end of exercise and what you have created.

The idea is to have one page for each of the micro-projects through the semester and then hand-in the pictorial as a finished example of what you have done.



Unity

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Unreal

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SECOND BLOCK – CAPTURING REALITY

Here you learn about creating your own model/s that you will use throughout the course. Test out traditional photogrammetry, 3D scanning with your phone, creating a Gaussian Splatting or Nerf visualization. {Claus Help here does it make sense???? Are the hand-ins good?}

As a second part you will learn how to estimate the lighting and implement image-based lighting. How to delight the captured object, to create a clean albedo texture that can be lit using virtual lights.

Here you can use again whichever engine you prefer Unreal or Unity or another that you are more comfortable with. Show us your step by step work in creating your assets and preparing them for use further. How you captured light, etc. Did you create different lit scenarios or tested out multiple way to capture your 3D model. Maybe you decided to compare a hand-modeled model vs one captured from the real-world.

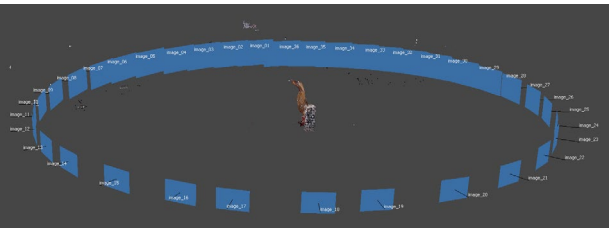


Producing your 3D model/s

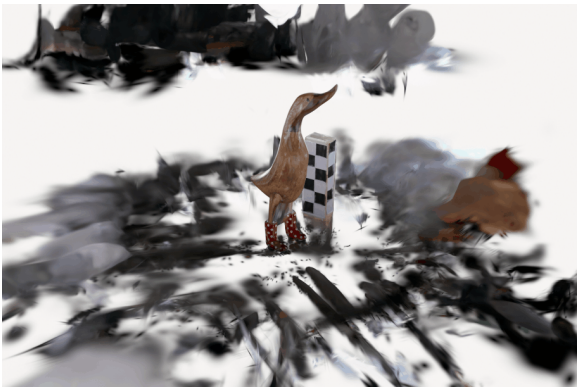
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Estimating Lighting

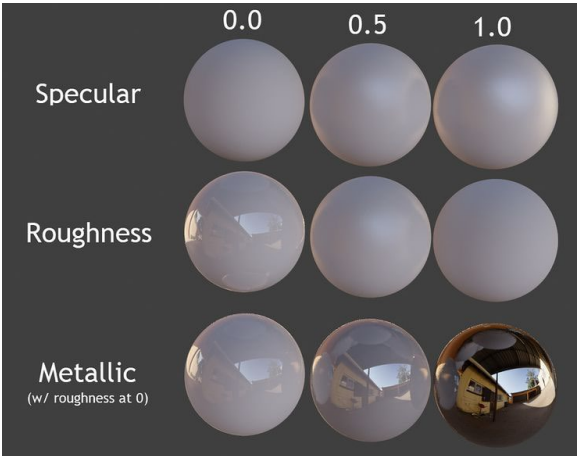
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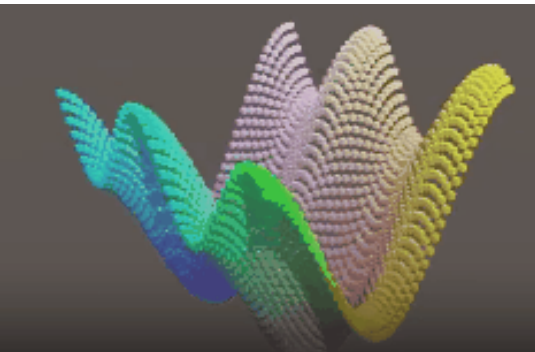
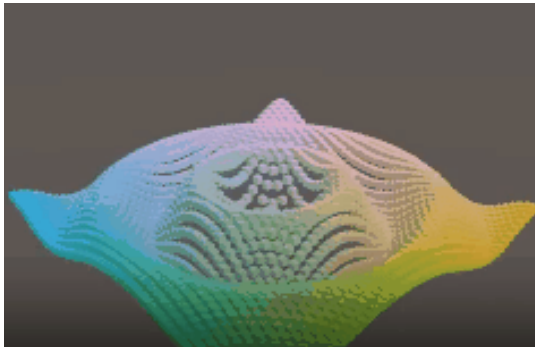
THIRD BLOCK – MATH FOR GRAPHICS

In this chapter we learn the core math principles useful for computer graphics. These are affine transformations in 2D and 3D, projections, rotations, matrix algebra, Euler angles, Quaternions. Together with interactions in space with lines, planes and polygons and calculating their parametric equations.

{Olga Help here does it make sense did I forget something???? Does the hand-in make sense below?}

As a hand in for this chapter use the created meshes and present interesting transformations and interactions built from scratch between them and other 3D objects. Give a short explanation what we are seeking in image, code or gifs. Let the images speak.

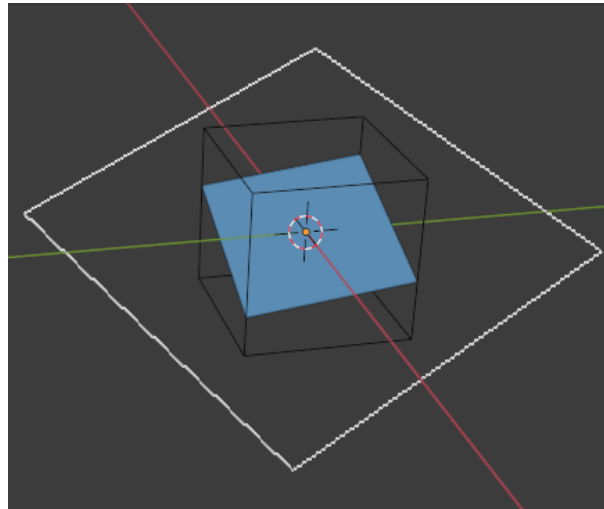
Transformations



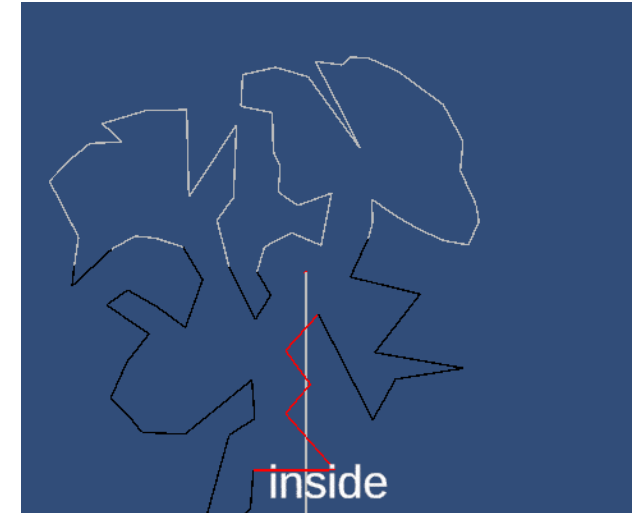
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Interactions and Parametric Representations

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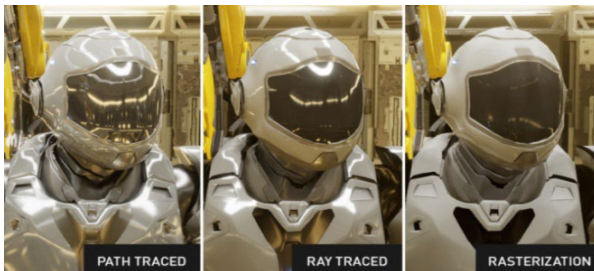
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FORTH BLOCK – LIGHTING, BRDF, RAYTRACING

In this chapter we go in detail about how light and matter interact – what happens in the real world and how we can transfer these complex phenomena to the more constrained graphics environment. We will go in detail about Bidirectional Reflectance Distribution Functions (BRDF), as well as different ways we can do raytracing, raymarching, pathtracing, and more. We will discuss what is high dynamic range and how we can use it to capture more information about light. {Claus Help here does it make sense???? Does the hand-in make sense did I forget something?}

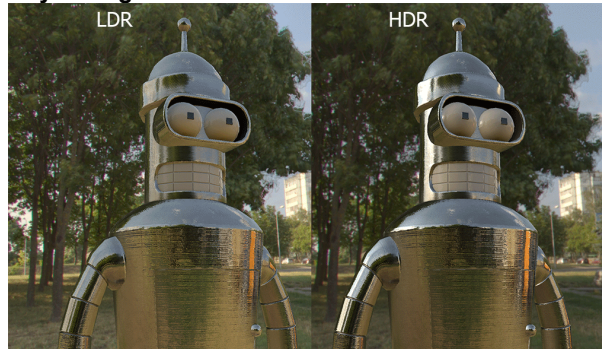
As hand-in you need to capture images of your mesh utilizing different ways to light it through raytracing, using physically based rendering, using captured HDRI maps, etc. Show side by side images of the same scene. Make scenes that are interesting from light stand point – indoor scenes with windows, multiple light sources, outside scenarios with different weather conditions. Show how lighting and shadows change based on time of day, etc. Give a short overview of what you have tried, how you have setup your scene, how you have captured maps, how much time did you need to render.

Raytracing



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Raytracing

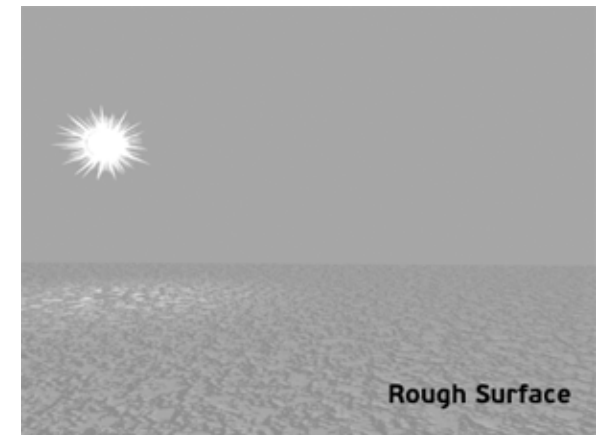


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Raytracing

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BRDFs



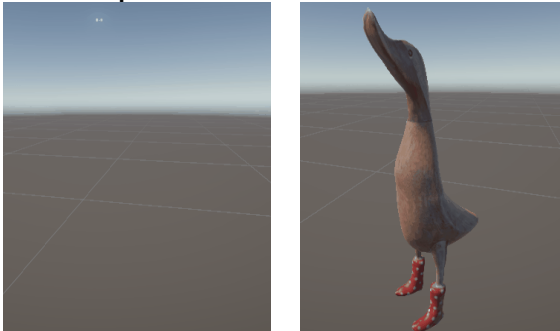
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FIFTH BLOCK – GRAPHICS PIPELINE AND SHADERS

In this chapter we will go through the traditional rasterization graphics pipeline, as it is still an industry standard. We will then go into how we can create shaders using Unity's Shader Graph. As a second point we will also learn how to build upon this by programming shaders through code using HLSL/Cg in Unity. The same things can be used in Unreal through the material designer.

As a hand-in you should create shaders that can manipulate your model, can change its vertices or add effects to it. If you want you can go one step further and try to create a compute shader to render a huge number of objects in screen, create camera shaders to add camera based effects, etc. You will need to try to make at least one of the shaders using coding. You will need to give a short explanation about how the shader is created, did you use any special textures or formulas.

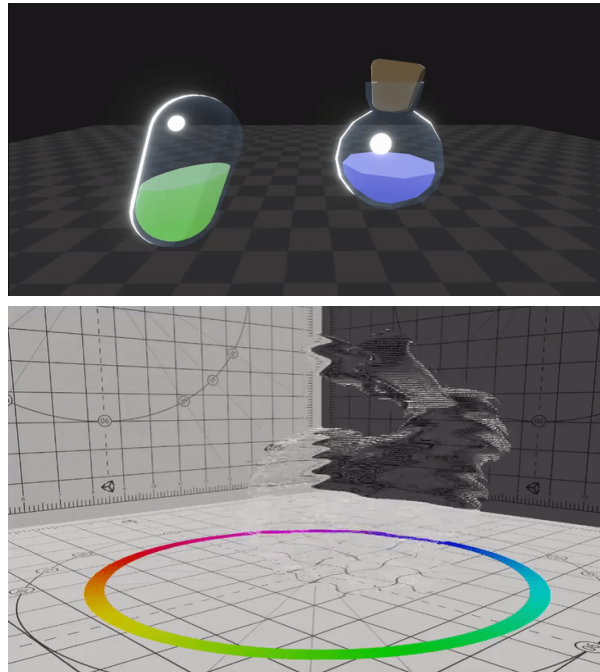
Shader Graph



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Programmed Shaders

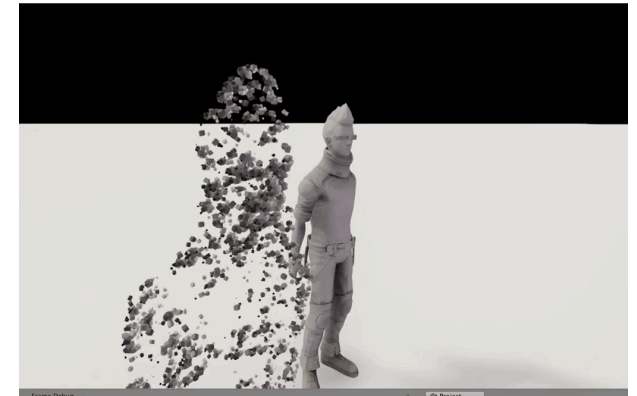
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Compute Shader, Camera Shader, Decal Shader, etc.

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FIFTH BLOCK – VIRTUAL PRODUCTION

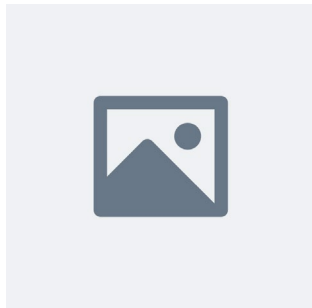
In this block we will discuss how computer graphics can be used as part of virtual production to blend real life people and props with virtual objects..... [Henrik help here](#) :P

{We need requirements for the assignment and maybe some visuals from ViZARTS maybe?} I will keep a lot of the lorem ipsum here plus some figure empty space.

What are hand-in requirements?

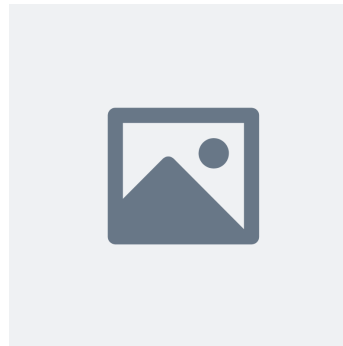
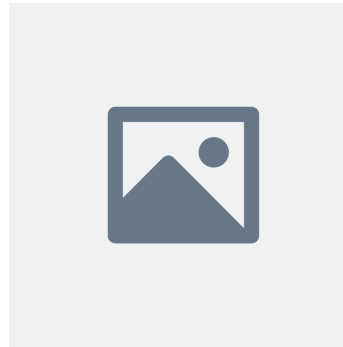
Virtual Production in Studio?

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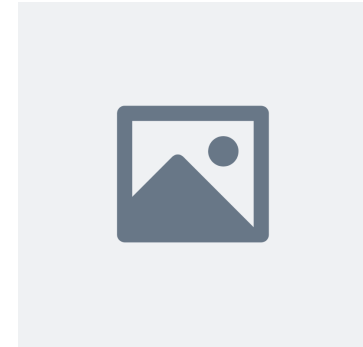
Virtual Production on a Low Budget

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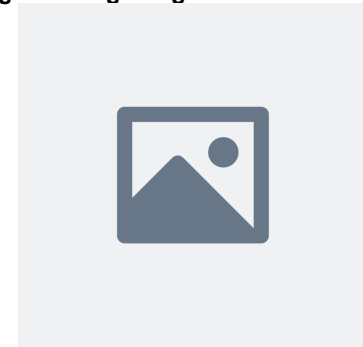


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Something I am forgetting???



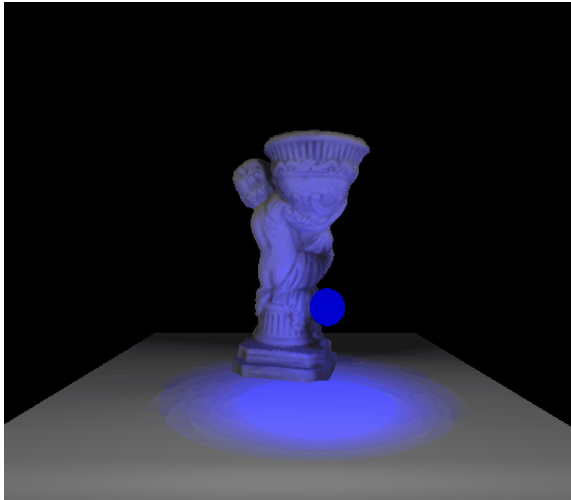
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SIXTH BLOCK – ANALYSIS AND VISUALIZATION OF MESHES

In this last chapter we will discuss how you can analyze your meshes, extract information from them, change them to make them smoother, smaller, segment them, voxelize them, among other things. We will also give a overview of python based libraries that are easy to setup and use for visualization of meshes, datasets, point clouds, where you can create beautiful plots for your semester projects or other courses.

As part of the hand-in you will choose one or more of the libraries that we discuss in python and try to analyze and visualize your meshes Or maybe you found another library or you want to try out something – then ask and we can figure out if there is a library for it. Extract interesting information about it, transform it, etc. Give a short overview of the different ways you have manipulated and visualized your meshes. Have you used interesting formulas, a method that is very useful, specific visualization style, etc.

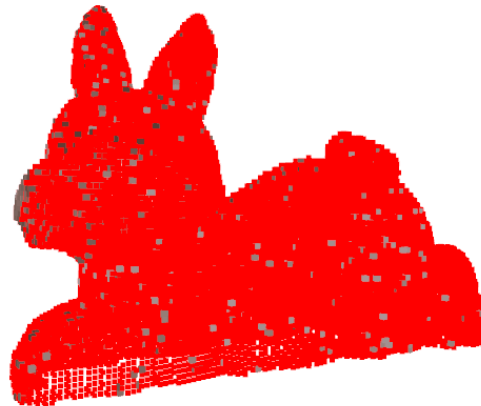
Mesh Visualization



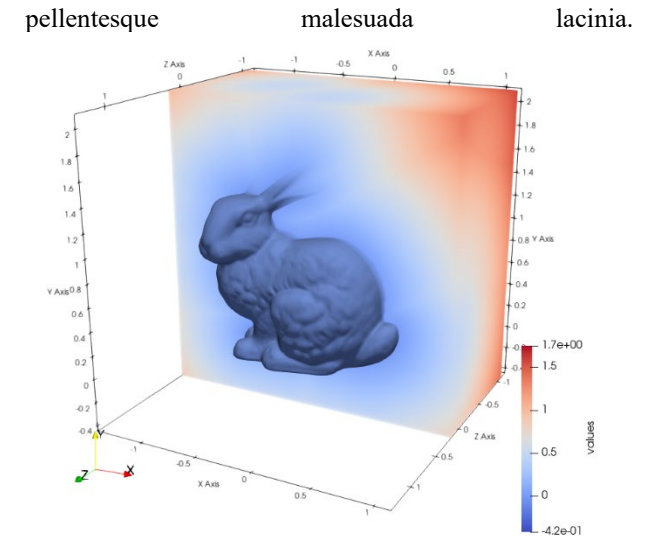
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Voxelization, Creating Point Clouds, using Signed Distance Fields (if you are brave enough)

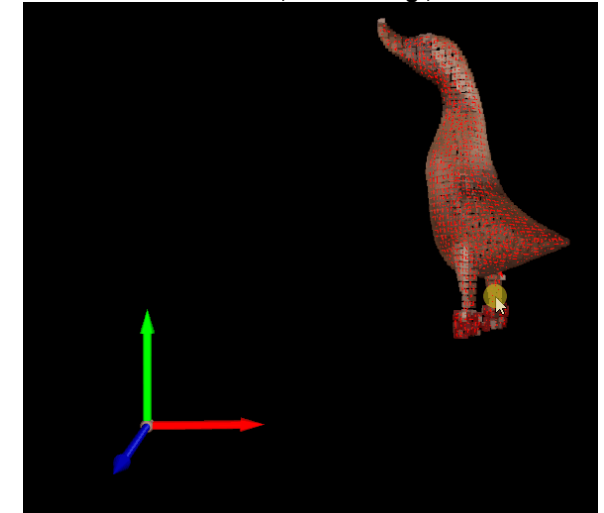
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Point Set Visualization, Smoothing ,etc.



REFERENCES AND CITATIONS

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REFERENCES

Here add any references to libraries, materials, assets or code that you have used from github, youtube, papers, articles, etc.

- [1] @_CHINOSAUR. 2014. VENUE IS TOO COLD. #BINGO #CHI2016. Tweet. (1 May, 2014). Retrieved February 2, 2014 from https://twitter.com/_CHINOSAUR/status/461864317415989248
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